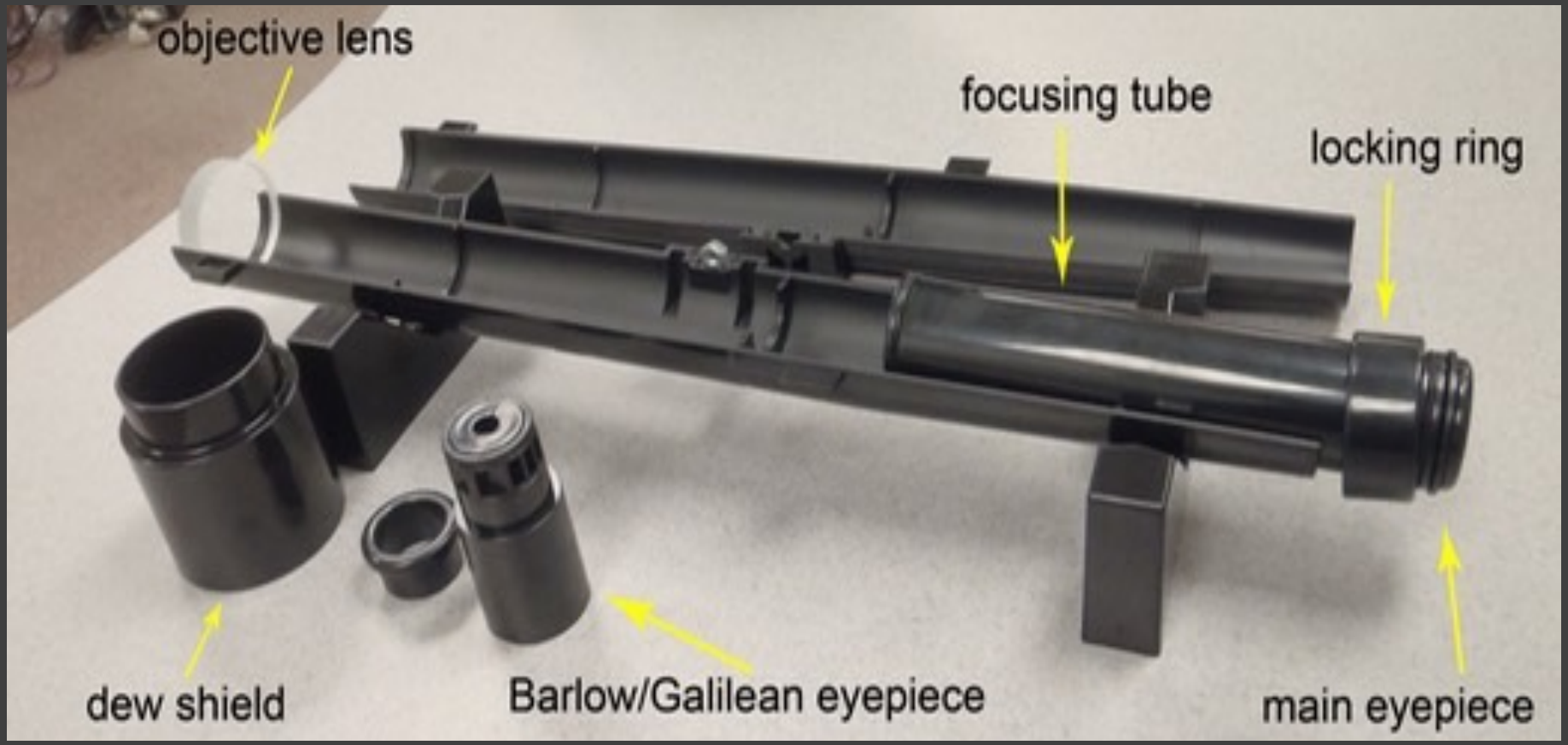


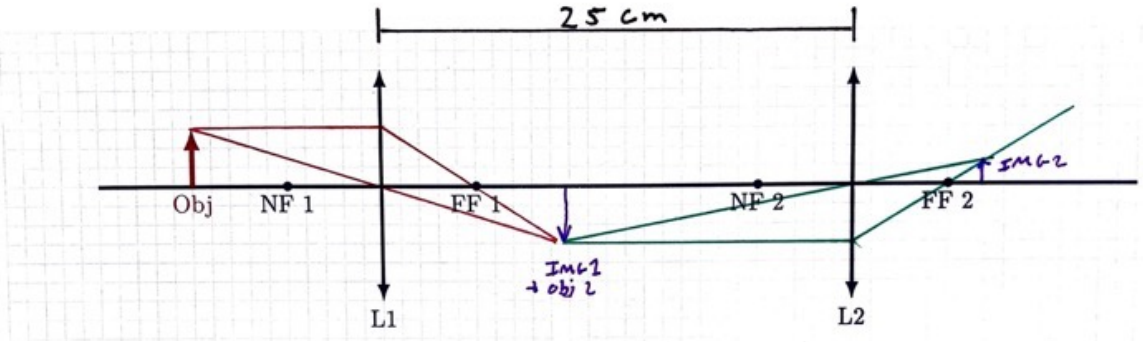
Optical Devices

What happens when you have multiple lenses/mirrors?

The image of the first Lens becomes the object of the second

And the system magnification: $M_{\text{sys}} = M_1 M_2$

Ex. Two converging lenses each with a focal length of 5cm are separated by 25 cm. A 3 cm tall object is placed 10 cm from the first lens. Where are the first and second images? What is the system magnification?



IMG 1: $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$\frac{1}{5} = \frac{1}{10} + \frac{1}{d_i}$

$d_i = 10\text{cm}$

$M_1 = -\frac{d_i}{d_o} = -\frac{10}{10} = -1$

Now distance w.r.t. L_2

So $d_i = 10\text{ cm}$

$\rightarrow d_o = 15\text{ cm}$

IMG 2: $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$\frac{1}{5} = \frac{1}{15} + \frac{1}{d_i}$

$d_i = 7.5\text{ cm}$

$M_2 = -\frac{d_i}{d_o} = -\frac{7.5}{15} = -0.5$

$M_{sys} = M_1 M_2 = (-1)(-0.5) = 0.5$

Ex. A 2 cm object is placed 5 cm from the first lens in a two lens system. The first lens is a convering lens with f_1 = 2 cm. The second lens is a converging lens with f_2 = -4 cm. The two lenses are 10 cm apart.

A. Use ray tracing to locate the first and second images

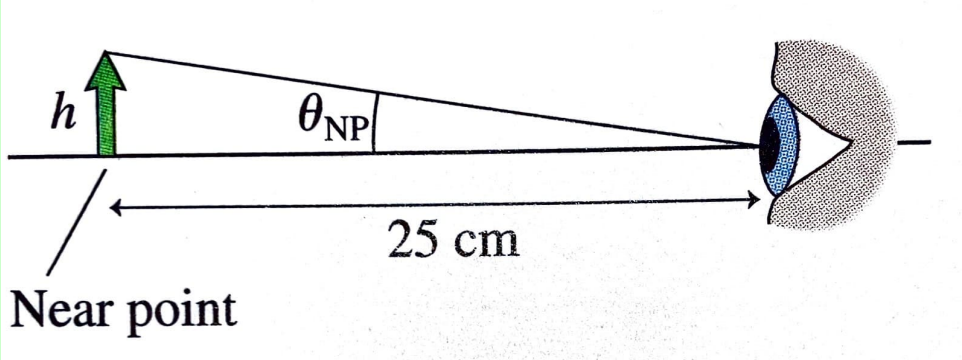
B. Use the thin-lens equations to solve for the first and second image distances

C. Find the magnification provided by the first and second lenses respectively

D. What is the overall system magnification?

Near point:

When you look at objects the angle they **subtend** gets Larger and larger as you bring the object closer. But there Is a max angle as you cant focus on anything past your nearpoint (NP), which is about 10 in or 25 cm for most people.



Demo: Measure your NP

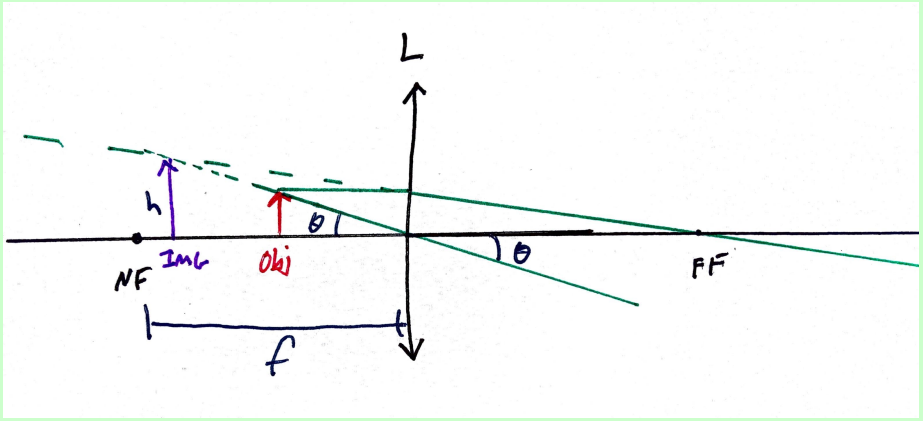
$$\theta_{NP} = \frac{h}{25\ \text{cm}}$$

Magnifying Glass:

Object is inside focus of converging lens



How does this even work?



The angle subtended by the image is approximately:

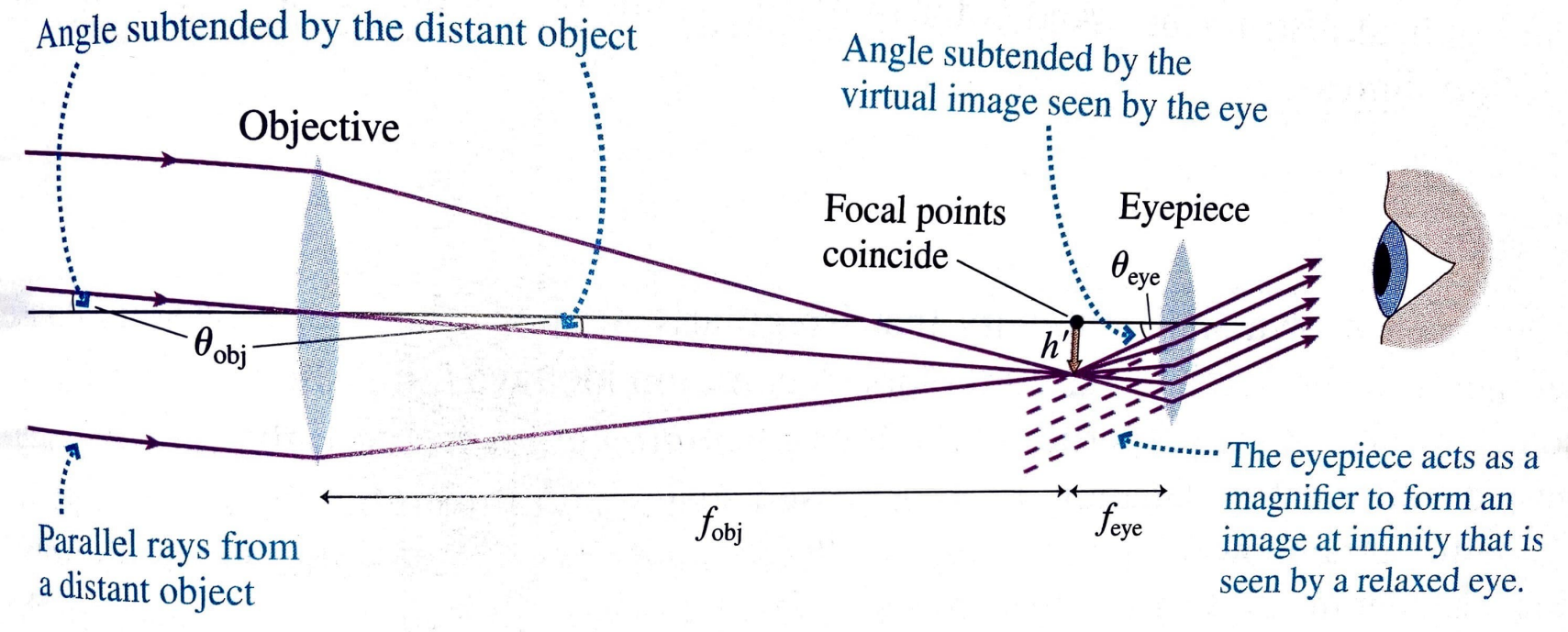
$$\theta = \frac{h}{f}$$

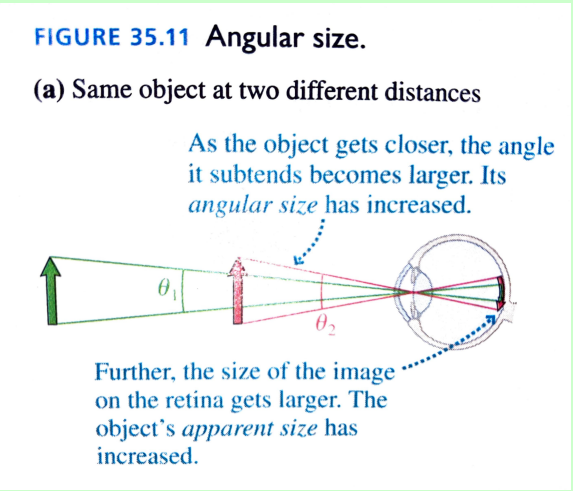
The angular magnification is thus:

$$M = \frac{\theta}{\theta_{NP}} = \frac{h/f}{h/25} = \boxed{\frac{25\text{ cm}}{f}}$$

Ex. What is the magnification of a 5 cm focal length magnifying lens?

Telescopes: Two converging lenses separated so that their two focuses overlap





Angular Magnification: $M = \dfrac{\theta_o}{\theta_e} = \boxed{-\dfrac{f_o}{f_e}}$

$\tan\theta_o = -\dfrac{h}{f_o}$ (negative b/c inverted Img)

$\theta_o \approx -\dfrac{h}{f_o}$ (small angle approx.) $\quad \theta_e \approx \dfrac{h}{f_e}$

Ex. Rays from a distant star come in 0.1 rad off the OA. If the objective has a focal length of 20 cm, and the eyepiece has a focal length of 4 cm..

A. How far apart should you place the two lenses to make a telescope?

B. Trace the rays through the telescope

C. What is the angular magnification of the telescope?

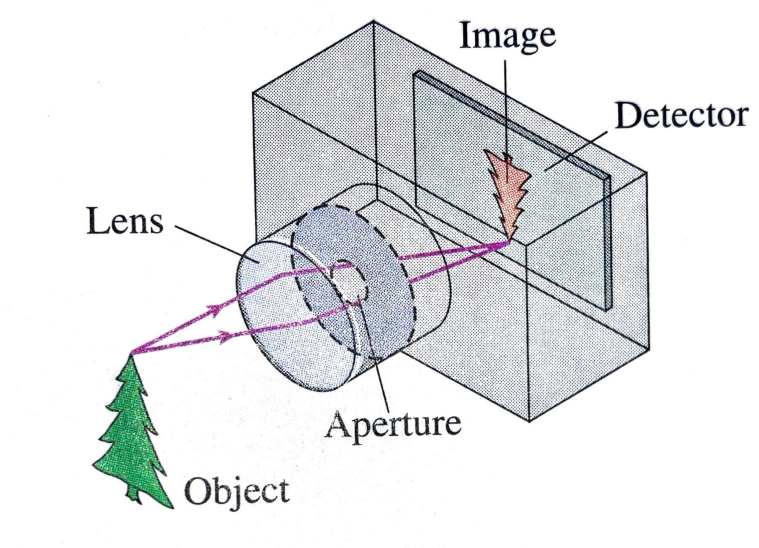
D. How big will the angle θ_e subtended in the eyepiece be?

Cameras: Lens creates an image on a detector that can be moved in or out

f = fixed

$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$ ← adjust this. (called focusing)

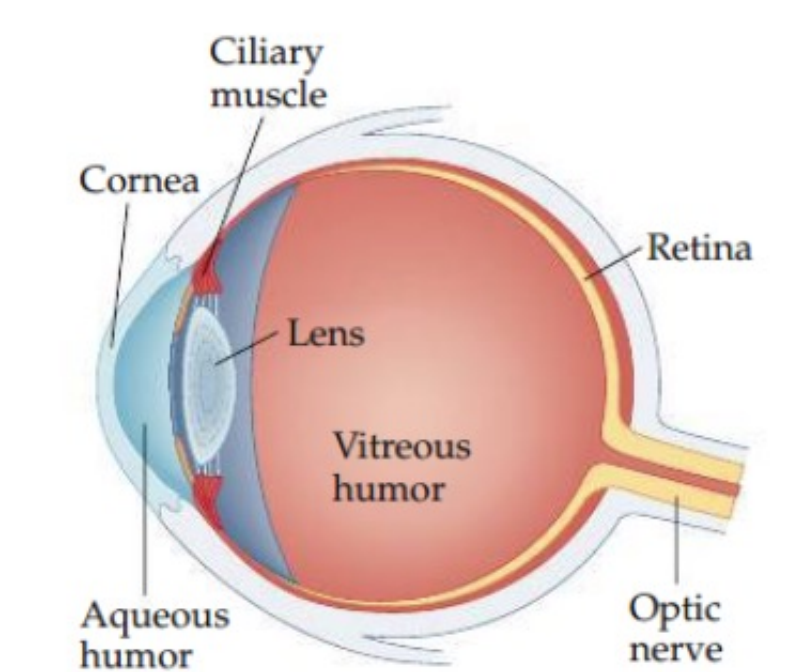
If this changes

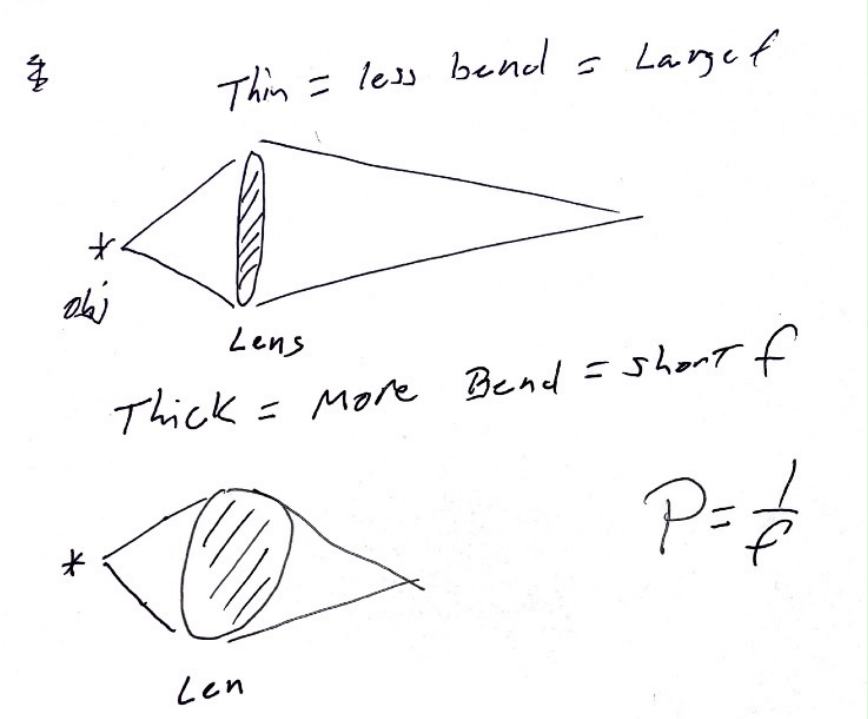


Ex. 50 mm lens means that if d_0 = infinity, then film should be 0.05 m behind lens.
What if object is closer than infinity?

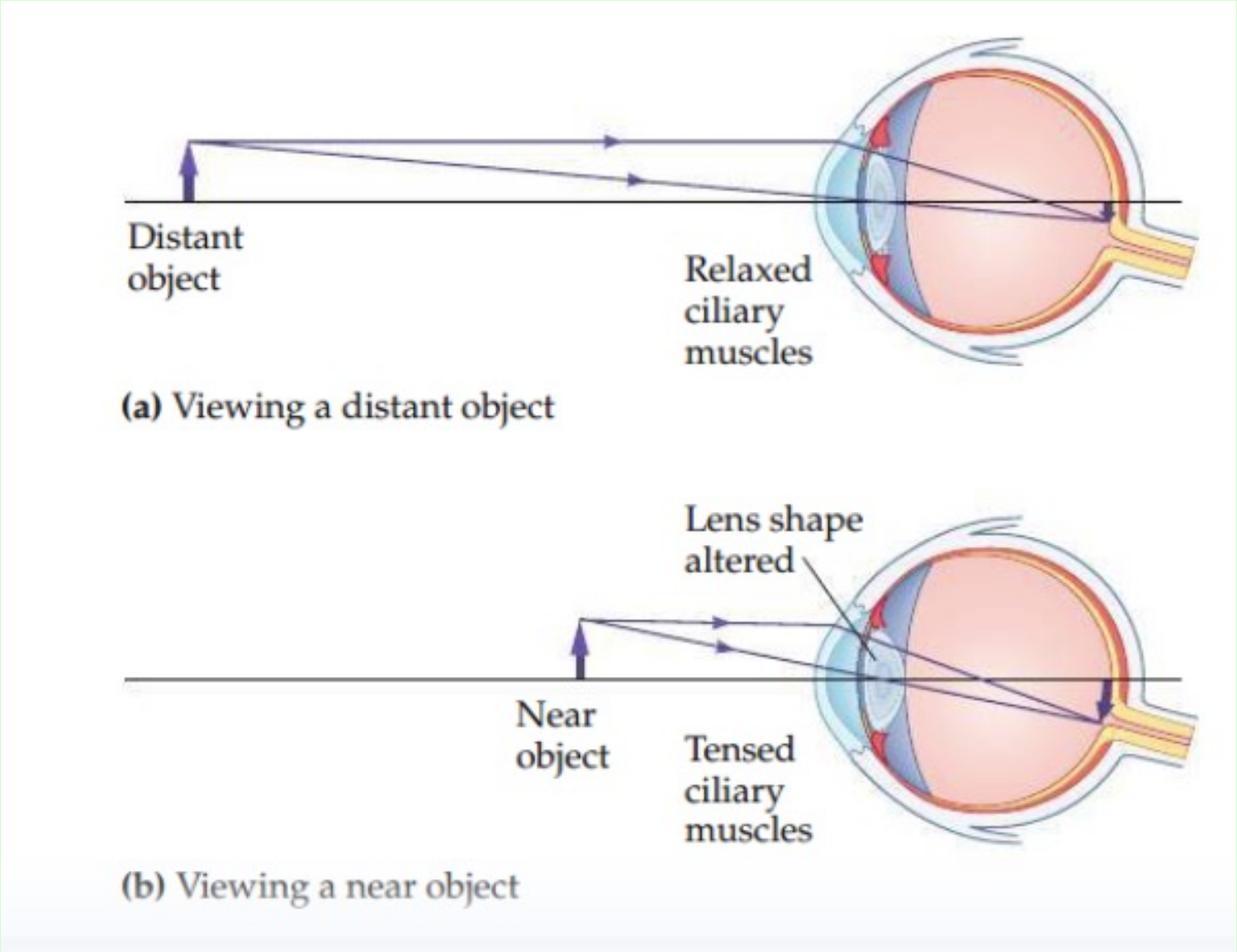
Eyes:

Thick lens have more power. Will divert light rays path more

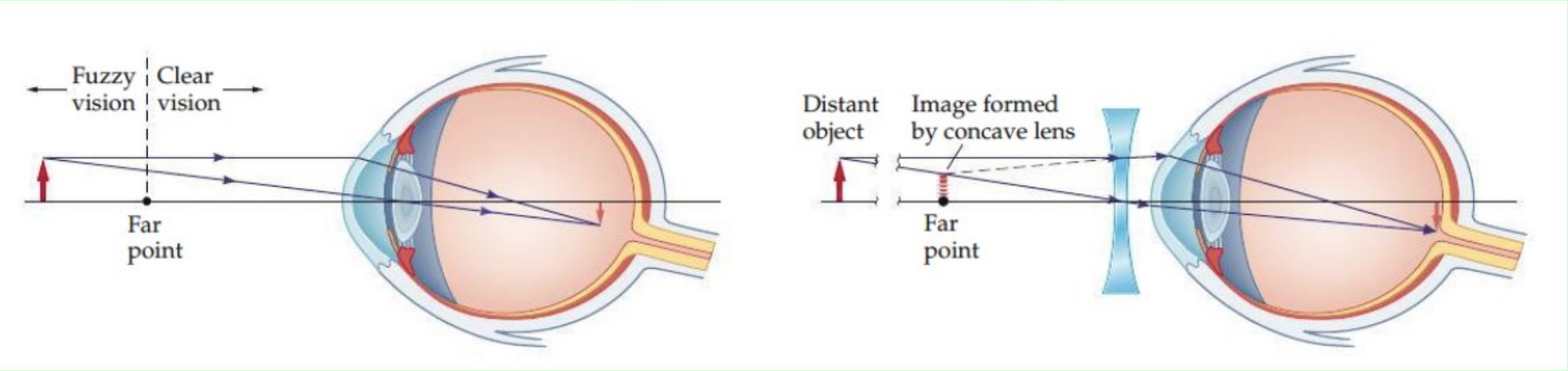




Normal Vision:



Myopia



Lens is too fat even at rest.
Even really far objects focus before retna

Diverging lenses spread out incoming Rays even more, to bring object to Focus on retna